Who We Are

- **Improved Fishing Device Controller**
- Ohm++
- Sponsored by Quality of Life Plus, partnered with River Deep Foundation

Garrett Hurd

- Electrical Engineer major
- Born and raised in the North Bay of California
- Admin Roles:
  - ✓ Team Lead
  - ✓ Schedule Manager
  - ✓ Backup for Assignments Manager
- Engineering Roles:
  - ✓ Power Management
  - ✓ Backup for System Architect
  - ✓ Backup for Analog Circuit Design

Tyler Martin

- Electrical Engineering Major
- Born in Hawaii, Raised in Colorado
- Admin Roles:
  - ✓ Customer Manager
  - ✓ Backup Communications Manager
  - ✓ Backup Moral Manager
- Engineering Roles:
  - ✓ Analog Circuit Designer
  - ✓ Test Engineer
  - ✓ RF Lead
  - ✓ Backup Mechanical Engineering Lead
  - ✓ Backup Power Management Architect

Pedro Mejido

- Electrical Engineer major
- Born and Raised in Texas
- Admin Roles:
  - ✓ SCRUM Master
  - ✓ Backup Budget Manager
- Engineer Roles
  - ✓ PCB Lead
  - ✓ System Architect

Felipe Tamayo Tapia

- Electrical and Computer Engineering major
- Born in Denver, Colorado
- Admin roles
  - ✓ Morale manager
  - ✓ Schedule manager
- Engineer roles
  - ✓ Embedded systems
  - ✓ Software engineer

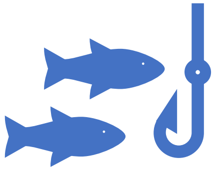
Vedant Shah

- Electrical and Computer engineering major
- Admin Roles:
  1) Budget Manager
  2)Logistic Manager
  3) Backup Team lead
- Engineering Roles:
  1) Embedded Systems
  2) Software engineer
  3) User Interface

Travis Wood

- Electrical Engineer major
- From Washington State
- Admin Roles:
  - ✓ Communications Manager
  - ✓ Assignments Manager
  - ✓ Backup for Scrum Master and Customer Manager
- Engineering Roles:
  - ✓ User Interface
  - ✓ Mechanical
  - ✓ Backup for PCB and Testing

Product objectives and purpose of the system



Intuitive, fun, and easy to use fishing rod for people with spinal cord injuries and traumatic brain injuries (TBIs).

Used by patients of Craig Hospital during recreational therapy outings

Important behaviors and capabilities:

Easily used by spinal cord and TBI patients

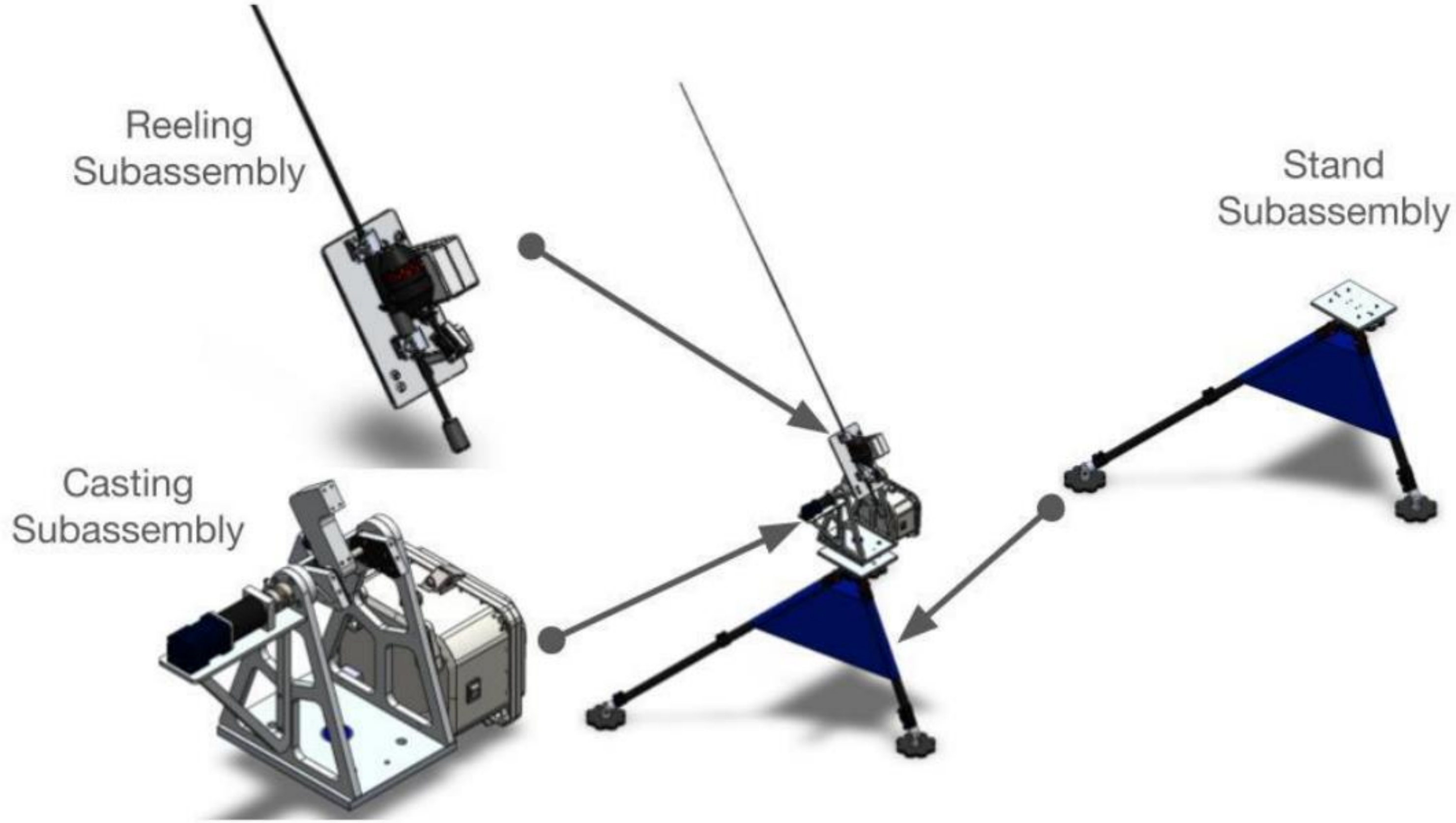
Intuitive design, no manual required

Two hours of battery life

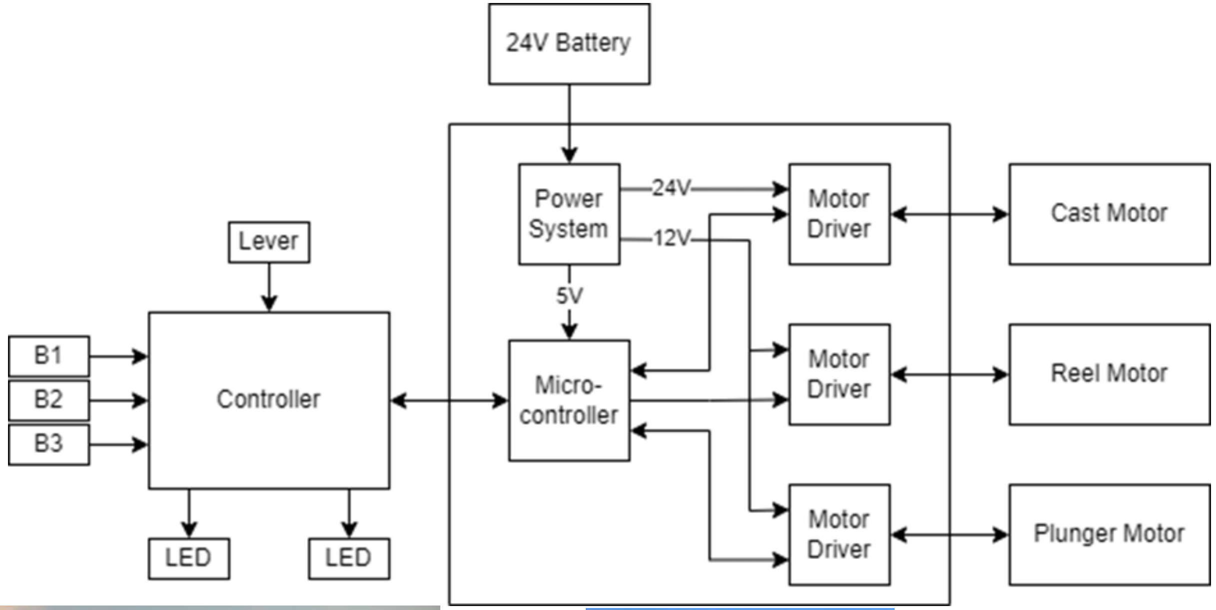
<u>Our number one priority is building the best user experience!</u>

Overall Product Design

Overall Product Framework



Overall Product Design



24V battery Input
Around 2 Amps of current draw

Worst case scenario:
20 casts/per hour

Optimal case
~30 casts/per hour

Inputs: 3 Buttons, Lever
Outputs: 3 motors, LEDs

Technology options to achieve your product requirements

Microcontroller option: ATMEGA 328

Why:

- We are very familiar with it
- It’s relatively inexpensive
- It’s easy to use

Highest risk components: Stepper Motor, Stepper Motor Driver

Why:

- Unfamiliar with how it works
- Complex system

Technology option	Maturity risk	Complexity risk
ATMEGA 328	Low	Low
LEDs	Low	Low
Lever	Low	Low
Buck Converters	Low	Medium
Buttons	Low	Low
Stepper Motor	Low	Medium
Stepper Motor Driver	Low	Medium
DC Motor Driver	Low	Low
Speed Sensor	Low	Low
PID Speed Compensator	Low	Medium
Arduino IDE	Low	Medium
FreeRTOS	Low	Medium

Software Development Plan (FreeRTOS)

- Free Real Time Operating Systems
- Capable of making 5-7 threads
- Consists of hardware and software timers
- Capable of performing Inter-task communication as the OS provides locks(aka mutex)and semaphores.

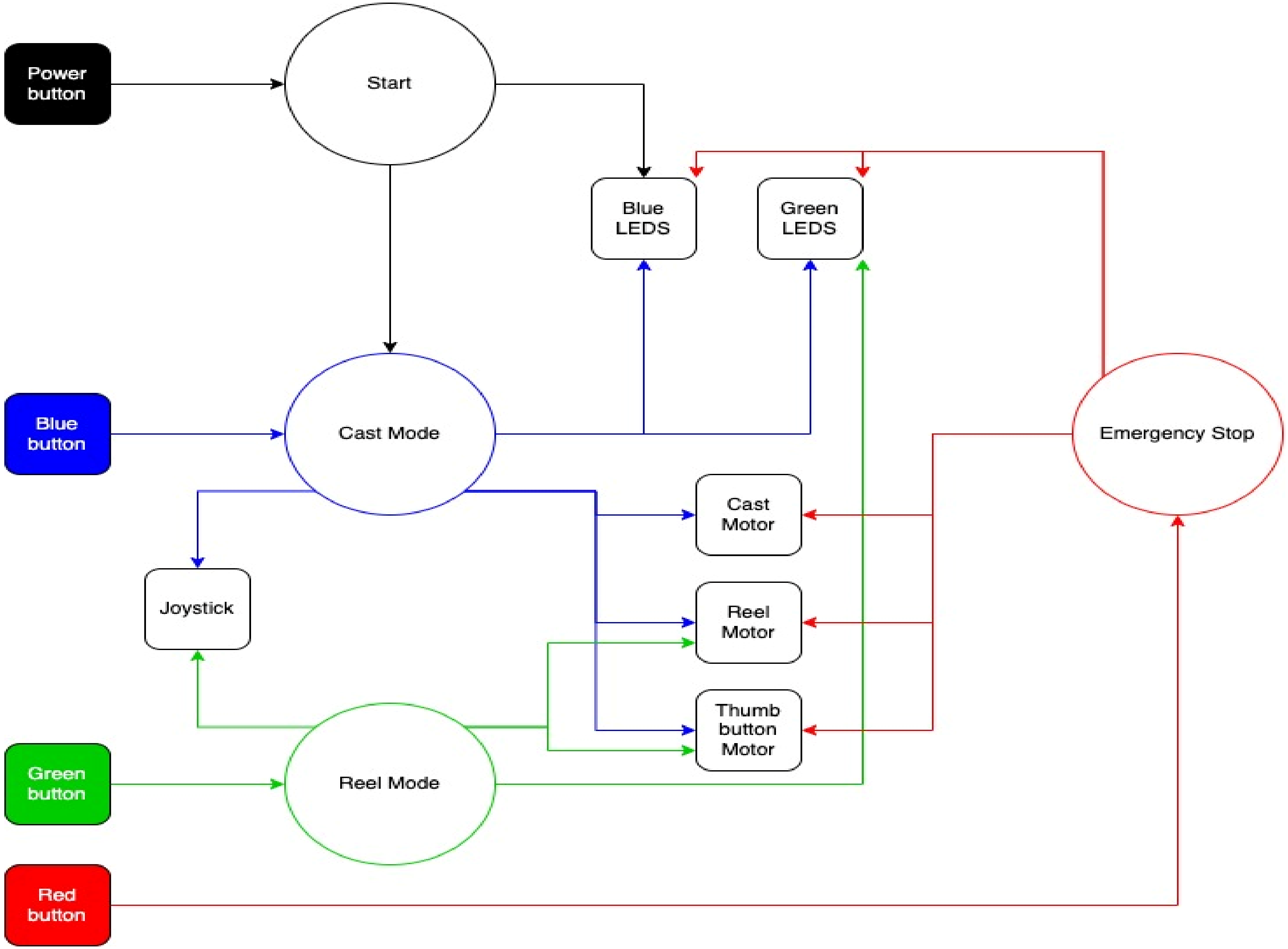
ARDUINO

IDE

freeRTOS

Operating System

Basic Flow chart of software using FreeRTOS



Potential Software Risks

- **Integration Risks:** Compatibility between FreeRTOS and Arduino libraries
- **Performance Issues:** Task prioritization and timing issues in FreeRTOS
- **Development Risks:** Debugging challenges (e.g., real-time debugging with FreeRTOS)
- **Mitigation Strategies:** Thorough testing, use of debugging tools (like serial monitors), fallback mechanisms

Project Risks, Test Plans and Mitigation Strategies

- CONTROLLER MUST BE INTUITIVE
  - ✓Sponsors number one concern and hardest to test
  - ✓Feedback from large sample size on a variety of designs and inputs
  - ✓Back up plan: make multiple controllers so that sponsor can use patients favorite
- Power Management (parasitic current draw)
  - ✓Concerned about the battery life draining very quickly
  - ✓Measure the output current from the battery while operating the fishing rod
  - ✓Find ways to mitigate unnecessary power loss
- How much will we keep from the previous design and how much will we replace
  - ✓Have to consider the maturity risk, complexity risk, time risk and financial risk of replacing components
  - ✓For example: do we replace the thumb button motor for a spring return solenoid

What is the plan for phase 1 (Proof-of-Concept of the system Design):

- Some of the key features:
  - ✓Current motors and platform
  - ✓3D printed controller
  - ✓Off the shelf motor drivers
  - ✓Will use an Atmega328 mounted to a custom PCB
- Fishing rod will have a variable casting distance and reel in speed along with an emergency stop button
- Function mostly the same as the final project without some of the quality assurance features
  - ✓Missing PID speed controller
  - ✓Power Management components

What is your test plan for phase 1?

- Cast distance
  - ✓Use a variety of weights on the line and back cast distance to measure cast distance to test reliability of casting
  - ✓Will look at how consistent the motor can cast by looking at the distribution of cast distance and the failure rate of casting over a very large sample size
- Test variable reel speed
  - ✓Set reel in speed and test a variety of input speeds with a variety of weights on end of line
- User interface
  - ✓Go to the business field and get feedback from random sample groups on how intuitive controller is, based on that feedback and time to up-and-running
  - ✓Will have a variety of controllers to get different insights into different designs

What is the plan for phase 2 (some Customization) and phase 3 (Final Hardware & Software):

Phase 2:

- Custom PCB in fishing rod and PID speed compensator
- Custom motor drivers

Phase 3:

- Tougher and water resistant (IP65) housing for controller
- Attach pad to controller
- Implementing stretch goals
- More fine tuning

Project Budget

Description	Quantity	Cost($)
Controller Casing	1	100
Buttons	3	90
Hall effect fader (Lever)	1	120
Motor drivers	3	50
Smart LED	~ 3	30
Batteries	1	300
PCB	1	40
Bite Tabs (Stretch goal)	1	80
Amphenol Connections	1	150
Total		**960**

Summary

Create an intuitive, rugged,and easy to use fishing controller for Craig Hospital patients with spinal cord injuries and TBI’s, that is capable of use for a recreational therapy fishing outing.

QL PLUS
Engineering Quality of Life for Injured Veterans

River Deep Foundation

Thank You!

Questions?